

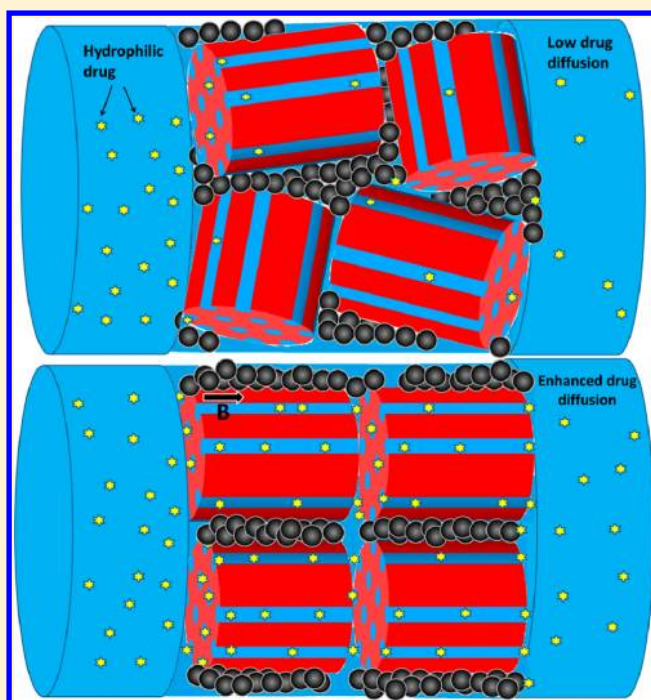
Controlling Anisotropic Drug Diffusion in Lipid-Fe₃O₄ Nanoparticle Hybrid Mesophases by Magnetic Alignment

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S Supporting Information

ABSTRACT: We present a new strategy to control the anisotropic diffusion of hydrophilic drugs in lyotropic liquid crystals via the dispersion of magnetic nanoparticles in the mesophase, followed by reorientation of the mesophase domains via an external magnetic field. We select a lipid reverse hexagonal phase doped with magnetic iron oxide nanoparticles and glucose and caffeine as model hybrid mesophase and hydrophilic drugs, respectively. Upon cooling through the disorder–order phase transition of the hexagonal phase and under exposure to an external moderate magnetic field (1.1 T), both the nanoparticles and the hexagonal domains align with their columnar axes along the field direction. As a result, the water nanochannels of the inverted hexagonal domains also align parallel to the field direction, leading to a drug diffusion coefficient parallel to the field direction much larger than what was measured perpendicularly: in the case of glucose, for example, this difference in diffusion coefficients approaches 1 order of magnitude. Drug diffusion of the unaligned reverse hexagonal phase, which consists of randomly distributed domains, shows values in between the parallel and transversal diffusion values. This study shows that modifying the overall alignment of anisotropic mesophases via moderate external fields is a valuable means to control the corresponding transport tensor of the mesophase and demonstrates that the orientation of the domains plays an important role in the diffusion process of foreign hydrophilic molecules.



INTRODUCTION

Lipid amphiphiles can form a variety of nanostructured lyotropic liquid crystalline (LLC) phases depending upon temperature,¹ composition,² and other variables.³ Recent studies show evidence of the presence of inverted liquid-crystalline phases during biological membrane fusion and the implications that this may have on membrane protein function,⁴ which may span from biomolecules to ion transport through nanochannels. Moreover, there is growing interest in using lipid-based nonlamellar phases such as bicontinuous cubic phases and columnar hexagonal phases in bioapplications and in the formulations of novel drug delivery systems.^{5,6} The unique potential of these systems has to be found in their biocompatibility, the well-defined pore structure that provides a diffusion pathway for controlled drug release, the ability to incorporate both hydrophilic and hydrophobic drug molecules, and their thermodynamic stability in excess water.^{7,8}

The transport characteristics of hydrophilic drugs within these liquid-crystalline phases strongly depend on the size of the water channels⁹ and the symmetry^{10,11} of the liquid-crystalline phases. For example, cubic phases in which the water channels are self-organized in 3D bicontinuous symmetry have 4-fold-faster diffusion than in columnar 1D hexagonal phases.^{10,11} In contrast, hexagonal phases can be used to produce molecular-size-selective aqueous nanofilters with greater control over nanoscale pore properties than with commercial polymer membranes.¹² However, one of the main limitations of this specific symmetry of mesophases is the low water flux associated with the random alignment of the hexagonal domains within the membrane. Therefore, an increase in the speed and control of the diffusion process of

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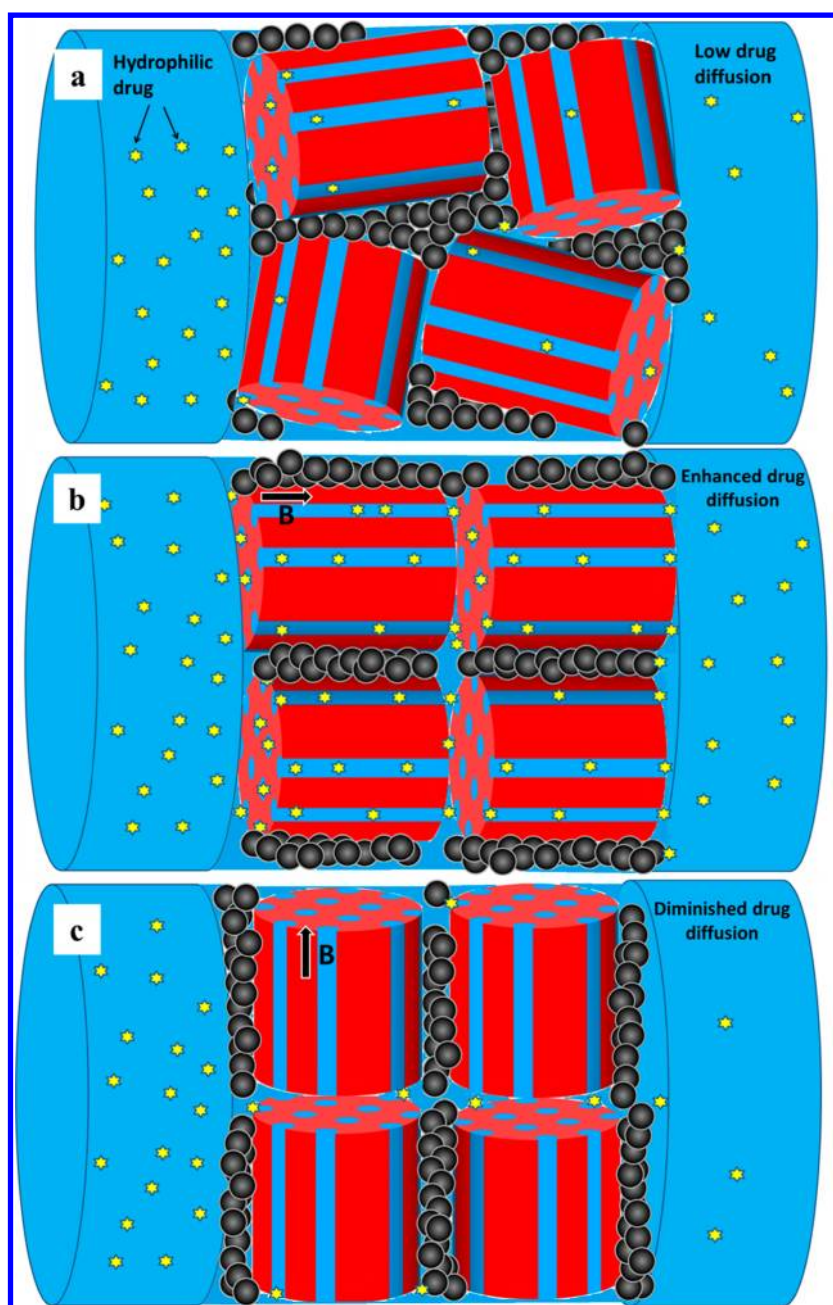


Figure 1. Schematic illustration of anisotropic drug diffusion in a columnar hexagonal liquid crystal/nanoparticle hybrid system. Drug diffusion in (a) a randomly aligned hexagonal phase, (b) a hexagonal phase aligned parallel to the magnetic field and diffusion direction, and (c) a hexagonal phase aligned parallel to the magnetic field and perpendicular to the diffusion direction.

the anisotropic pathways through the hexagonal water channels can benefit both drug delivery and membrane-related applications.¹³

Previous reports show that anisotropic ionic conductivity can be efficiently achieved by aligning self-assembled soft materials.^{14,15} Different alignment techniques such as a high magnetic field,¹⁶ electric field,¹⁷ shear alignment,¹⁸ and solvent casting¹⁷ have been employed to achieve anisotropic ionic conductivity through aligned self-assembled soft materials; however, all of the above reports have mainly focused on thermotropic liquid crystals^{14,18} or block copolymers,^{15–17} with only one exception in lyotropic supramolecular liquid crystals.¹⁹ Furthermore, all of these studies have dealt exclusively with proton and lithium ion transport, targeting technological

applications in the area of energy materials. The anisotropic diffusion of model drug molecules in a lyotropic columnar hexagonal phase has not yet been investigated, although this may have important implications in potential applications, not only in drug delivery formulations²⁰ but also in biosensors²¹ and biomedical materials.²²

The scarcity of data regarding anisotropic drug diffusion in lyotropic liquid crystals, together with exciting opportunities for these materials in drug delivery applications,^{5,6} provides strong motivation for original research in this area. In this context, the ability to control the orientational order in a reliable, noninvasive manner is highly desirable for controlled drug delivery. This remains challenging because conventional methods such as mechanical shear or electric field alignment

cannot be applied as a result of their invasiveness, issues of scalability, and dielectric break down. External magnetic fields of moderate intensity stand as a suitable candidate for alignment because of their low invasiveness, the possibility to apply them locally, and because the geometric constraints encountered with either mechanical shear or electric field alignment are missing.

In this letter, we show for the first time that aligned, inverted hexagonal columnar phases can be used to control the anisotropic diffusion of model hydrophilic drugs efficiently. This work builds on our recent research aimed at controlling the alignment of lyotropic liquid crystals by the confinement of magnetite nanoparticles within the grain boundaries of hosting mesophases²³ and subsequent alignment upon application of external magnetic fields. Although in our earlier work the magnetic alignment effect was demonstrated conclusively, its impact on final transport properties was not studied. Here we show that, following the same alignment strategy, the inverted hexagonal phase domains with their nanometric water channels aligned parallel to the columnar axis (and magnetic field director) have a diffusion coefficient of the hydrophilic drug that can approach a 1 order of magnitude increase compared to what is measured in directions orthogonal to the field. The main idea behind the control of drug diffusion in the hybrid hexagonal mesophase is depicted in Figure 1. Generally, in bulk samples hexagonal domains are oriented randomly because of the entropically favorable conditions. Drug molecules within a mesophase domain diffuse primarily through the water channels, and then they can hop through the domain boundaries to the next domain. This nonuniform alignment results in a considerable decrease in drug diffusion because of the discontinuous pathway encountered by the diffusing drug molecules through the water channels and the largely unfavorable orientations of the channels (Figure 1a). In the presence of an applied magnetic field and upon cooling through the disorder–order transition temperature (DOT), both the nanoparticles and the hexagonal domains mutually align along the magnetic field, enhancing the diffusion in this direction (Figure 1b). This does enhance the drug diffusion in the aligned direction because more and more drug molecules can diffuse in an unrestricted way along the hexagonal water channels, which, being favorably aligned, tend to offer a continuous pathway to the diffusion of hydrophilic molecules through the mesophase. Similarly, a decrease in drug diffusion can be obtained by aligning the hexagonal domains perpendicular to the diffusion direction, thus further decreasing the drug transport (Figure 1c). This strategy introduces a simple, facile pathway to the control of the anisotropic transport tensor in inverted lipidic mesophases, opening new opportunities in the sustained release of drugs and active ingredients.

MATERIALS AND METHODS

Sample Preparation. Dimodan U/J (Danisco, Denmark), an industrial grade of monolinolein, was selected as a neutral lipidic surfactant and used as received. Distilled deionized (Milli-Q grade) water was used for the formulation of the hexagonal phase. The phase behavior of water/Dimodan U/J has been extensively investigated in our earlier work.² In the present case, the water component was replaced by an aqueous suspension of 10 w/w% magnetite nanoparticles. According to the previous report, to create columnar inverse hexagonal phase at room temperature, Dimodan was mixed with Limonene.²⁴ Lipid blended with 15 wt % Limonene (Sigma-Aldrich) was prepared by mixing 2.55g of Dimodan with 0.45g of

Limonene. The final hexagonal phase was prepared by mixing an 85 wt % lipid blend with a 15 wt % aqueous solution containing 10 w/w% Fe₃O₄ nanoparticles. For the diffusion studies, both glucose (Sigma-Aldrich) and caffeine (Sigma-Aldrich) were dissolved in water at a concentration of 1 wt %.

Methods. Small-angle X-ray scattering (SAXS) measurements were performed using a microfocussed X-ray source of wavelength $\lambda = 1.54 \text{ \AA}$ operating at 45 kV and 88 mA. The diffracted X-rays signal was collected on a gas-filled 2D detector. The scattering vector q was calibrated using silver behenate. Samples were sandwiched between two thin mica sheets and sealed with a O-ring, forming a sample with a thickness of ca. 1 mm. Measurements were performed at 37 °C, and samples were equilibrated for 30 min prior to measurements, and the scattering intensity was collected over 30 min and azimuthally averaged to yield a plot of 1D intensity versus scattering vector q . Polarized optical microscopy (POM) was performed under both cross and parallel polarized light using a Zeiss Axioskop 2 instrument. Small-angle neutron scattering (SANS) experiments were performed at the SANS II beamline of the SINQ station of the Swiss Neutron Source at the Paul Scherer Institute, Switzerland. The incident beam had a wavelength of $\lambda = 0.5269 \text{ nm}$, and the sample-to-detector distance was 1.2 m, corresponding to a q range of 0.4–2.4 nm^{−1}. The 2D scattering spectra were azimuthally averaged. The scattering intensity versus azimuthal angle spectra were used to calculate the order parameter. Diffusion studies were carried out in duplicate samples using the same method described in our previous paper,¹¹ and the magnetic alignment of the hexagonal phase/nanoparticle hybrid system was carried out in the diffusion chamber. The magnetic alignment of the hexagonal phase/nanoparticle hybrid system in the direction parallel to the diffusion chamber was done using a superconducting magnet (cryogenic, U.K.) with a uniform magnetic field of 1.1 T whereas the magnetic alignment orthogonal to the diffusion chamber was done using a Halbach permanent magnet with the same magnetic field of 1.1 T. In both cases, the mesophase was heated to above the order–disorder transition in the presence of a magnetic field for a total duration of 2 min, after which it was cooled again through the disorder–order transition. For the diffusion studies, the concentration of diffused glucose was determined by means of an optical rotatory dispersion device (ORD-Y02/15) mounted on a CD spectrometer (Jasco J-815). The measurements were performed at room temperature, and the ORD signal was acquired for each sample three times at a fixed wavelength of 350 nm; a series of known concentrations of glucose were prepared to construct a calibration curve, and the drug concentration was determined by interpolation. The concentration of diffused caffeine was determined by UV–vis spectroscopy (Cary-100 Bio UV–visible spectrophotometer).

RESULTS AND DISCUSSION

It is well established that nanoparticles smaller than the mesophase periodicity are confined within the water channels, swelling the mesophase periodicity,²⁵ whereas larger particles are expelled and can reside in the grain boundaries of the mesophase domains.^{23,26} Figure 2 shows the small-angle X-ray scattering (SAXS) pattern of the hexagonal phase in the presence and absence of magnetic nanoparticles with average size of 15 nm. Both diffractograms show three diffraction peaks with a ratio of 1:3^{1/2}:2 characteristic of the hexagonal mesophase. These diffractograms confirm that the introduction of magnetic nanoparticles does not disturb the formation of the hexagonal phase and leaves the lattice parameter of the doped phase (5.36 nm) and the original phase (5.38 nm) nearly unaffected, which corresponds to radii of water channels of 1.482 and 1.488 nm, respectively. This indicates that the nanoparticles are partitioned in the grain boundaries of hexagonal domains, which is consistent with their size, greatly exceeding that of the hexagonal-phase water channels.

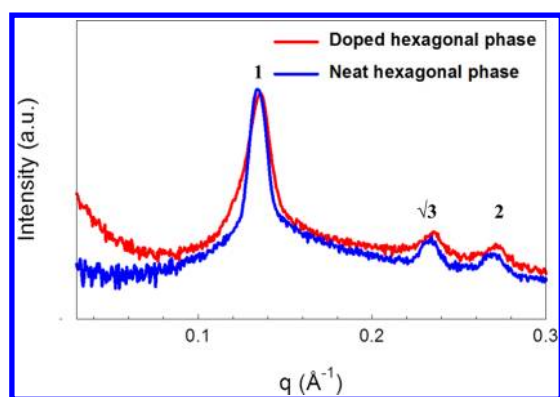


Figure 2. SAXS diffractogram for the neat hexagonal phase and the hexagonal phase loaded with 1.5 wt % magnetite nanoparticles. The identical lattice parameter for the neat and loaded hexagonal phases indicates that the nanoparticles are segregated within the grain boundaries of the mesophase domains.

The anisotropic behavior of the hybrid mesophase was assessed by small-angle neutron scattering (SANS) measurements. Figure 3a shows the 2D SANS diffraction pattern of the hexagonal columnar phase in the presence of nanoparticles but prior to its exposure to a magnetic field. The first Bragg's reflection of the hexagonal phase is azimuthally symmetric, showing isotropic alignment of the different columnar hexagonal domains. However, in the presence of an applied magnetic field of 1.1 T and upon cooling from the isotropic micellar phase to the hexagonal phase, a strong anisotropic pattern was observed as shown in the Figure 3b, with maximum intensities orthogonal to the magnetic field direction. This reveals that hexagonal domains are aligned with their director and their water channels parallel to the magnetic field direction. The integrated azimuthal intensity distribution of the first Bragg reflection is shown in Figure 3c. More quantitative information on the orientational order of the hexagonal domains can be extracted from the 3D order parameter (S_{3D}) using the following equation²⁷

$$S_{3D} = - \frac{\int_0^{\pi/2} I(\Theta) (3 \cos^2 \Theta - 1) \sin \Theta \, d\Theta}{\int_0^{\pi/2} I(\Theta) \sin \Theta \, d\Theta} \quad (1)$$

where $I(\Theta)$ is the scattering intensity distribution of the first Bragg reflection along the azimuthal angle Θ . The complete alignment of liquid-crystal domains would give values of S_{3D} approaching 1 (e.g., formation of a monodomain), whereas the

completely random alignment of liquid-crystal domains gives an S_{3D} value of zero, implying polydomain formation; values of S_{3D} in between quantify the orientational order in the LC. The 3D order parameter obtained accordingly was found to be $S_{3D} = 0.35 \pm 0.005$. Compared to the high 3D orientational order of the lamellar phase found in previous reports,^{23,28} the hexagonal phase shows a relatively smaller orientational order parameter because of the lower temperature window available for melting and renucleating the mesophase. For example, the hexagonal phase has an ODT of around 86 °C; the magnetic field could be applied only upon cooling from around 95 °C in order to preserve the integrity of the sample and to prevent possible water evaporation. Nonetheless, the macroscopic alignment gives access to the unidirectional spatial organization of the liquid-crystal domains, which can then be programmed in any desired direction by simply tuning the temperature and direction of the external magnetic field.

We then proceeded further to assess the relevance of the observed anisotropic orientation to the transport properties of hydrophilic drugs within the mesophase. Diffusion studies of the hydrophilic model drug glucose were carried out in a homemade experimental setup, following the procedure described in our previous report.¹¹ We measured the glucose diffusion in the hexagonal phase/nanoparticle hybrid system with macroscopically oriented water channels in different directions by the application of a 1.1 T magnetic field, as discussed above. The mesophase symmetry before and after the diffusion experiments was also investigated by SAXS (Supporting Information) to confirm that the symmetry of the mesophase is preserved for the entire duration of the diffusion experiments. Figure 4a illustrates the anisotropic drug diffusion in the hexagonal phase measured during the time in the receiving chamber. The drug diffusion in the mesophase when the water channels are aligned parallel to the magnetic field ($D_{\parallel}$) is significantly higher than that in the perpendicular direction ($D_{\perp}$), while the unaligned samples (D_x) show drug diffusion, which is intermediate between them. It should be noted that a slight variation in the alignment of hexagonal domains can create a noticeable variation in the diffusion profile of the drug in various samples. This effect is more pronounced in the samples aligned parallel to the columnar axis, and the signature for such diffusion behavior is found in the initial higher standard deviation of the diffusion of the horizontally aligned hybrid mesophase. The relatively small reduction in diffusion observed for the vertically aligned hexagonal phase compared to that in the unaligned hexagonal phase is expected to arise from diffusion through grain boundaries, which settles a

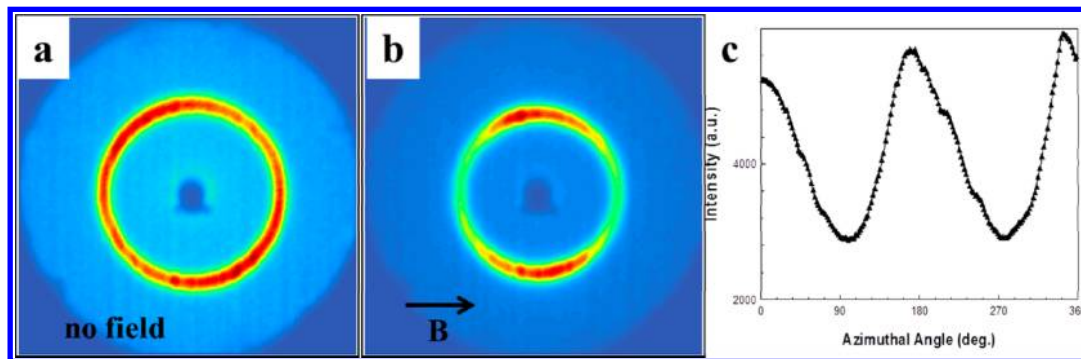


Figure 3. Two-dimensional SANS diffractogram (a) in the absence of magnetic field and (b) with the 1.1 T field applied as indicated. (c) Azimuthal intensity of the primary Bragg peak for the magnetically aligned sample.

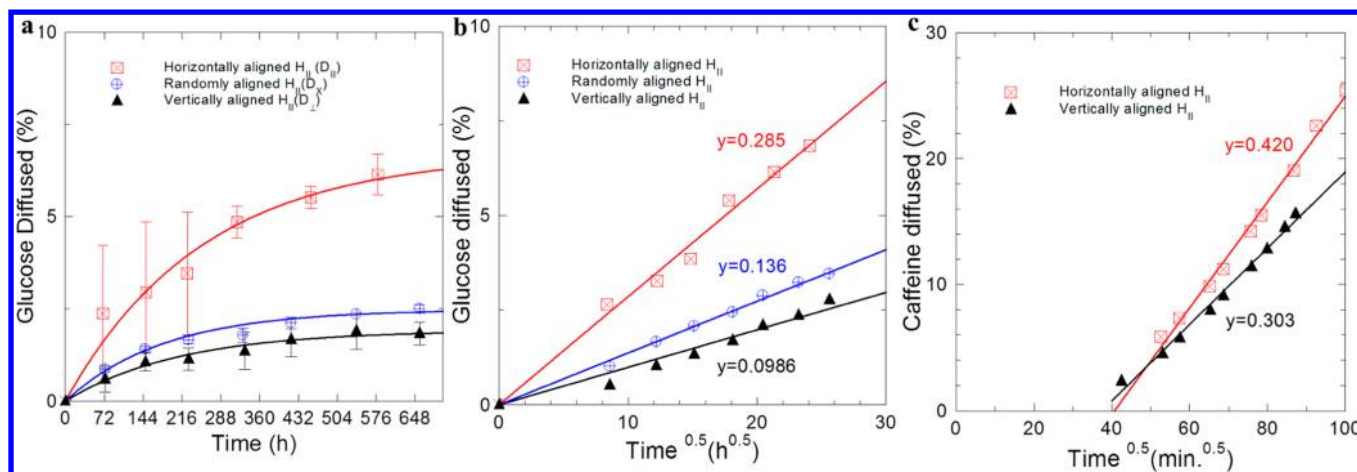


Figure 4. Diffusion behavior of hydrophilic model drugs through the hexagonal phase/nanoparticle hybrid system with different macroscopic alignment at 37 °C. (a) Percentage of glucose that diffused through the hexagonal phase plotted against time. (b) Percentage of glucose that diffused through the hexagonal phase plotted against the square root of time. Insets give the slope obtained from glucose diffused = slope(time) $^{0.5}$. (c) Percentage of caffeine that diffused through the hexagonal phase plotted against the square root of time.

lower boundary limit for the diffusion cutoff and partially smears the suppression in transport through the mesophase.

A more quantitative assessment of the anisotropic diffusion can be achieved by comparing the diffusion coefficients in the different directions. Because the setup that is used presents a lag time for the drug to diffuse from the donor to the receiving chamber, the exact determination of the diffusion coefficient is not straightforward. Nonetheless, Figure 4b, which gives the drug diffusion profiles plotted as a percentage of diffused drug against the square root of time, demonstrates linear behavior (e.g., a Fickian diffusion process). Thus, the flow of glucose Q follows with time t the relationship $Q \approx (Dt)^{1/2}$, with D being the corresponding diffusion coefficient. The relative changes in the diffusion coefficients can then be simply determined by comparing the squares of the slopes in Figure 4b. This analysis shows that compared to the diffusion coefficient of the perpendicularly oriented domains, $D_{\perp}$, the randomly oriented domains possess a diffusion coefficient of $D_X = 1.9D_{\perp}$ and that the parallel oriented domains possess a diffusion coefficient of $D_{||} = 8.3D_{\perp}$. Thus, the diffusion coefficient of hexagonal domains aligned parallel to the diffusion direction is nearly 1 order of magnitude larger than in the orthogonal direction. We then set out to test how sensitive to the specific drug this diffusion coefficient anisotropy can be and we carried out analogous experiments on another hydrophilic drug: caffeine (Figure 4c). In this case, the diffusion is found to be much faster than glucose, and surprisingly, a much smaller ratio of diffusion coefficients is found ($D_{||} \approx 2D_{\perp}$), which highlights how the level of achievable anisotropy in diffusion can be dependent on the specific drug and its interaction with the mesophase.

The ratio of ionic conductivity previously reported between the parallel aligned and perpendicularly aligned hexagonal phase was very high¹⁸ (41); however, these reports were based exclusively on thermotropic liquid crystals and block copolymers, where the macroscopic alignment was achieved by high external fields.^{16,17} Strictly speaking about the lyotropic liquid crystals, the ratio of ionic conductivity reported was low¹⁹ (5.8). Furthermore, the present study is based on the molecular diffusion within the water channels, which is considerably more challenging than ionic diffusion as well as the self-diffusion of water in the mesophase,²⁹ reported

previously. Finally, the alignment technique proposed here to achieve unidirectional hexagonal domains makes use of moderate external magnetic fields. As mentioned earlier, the 3D order parameter obtained for the hexagonal phase in this study is relatively low, lower than for the lamellar phase, for which, however, a thermodynamically stable coexistence with excess water is not available. It can be anticipated that increasing the 3D orientational order parameter in the hexagonal domains will lead to increased aspect ratios in the anisotropic drug diffusion coefficients.

CONCLUSIONS

We have demonstrated that in presence of a moderate magnetic field anisotropic diffusion of hydrophilic drugs in reverse columnar hexagonal phases can be obtained by aligning the domains of the mesophase in a suitable direction. When the hexagonal phase, doped by magnetic nanoparticles, is exposed to an external magnetic field of 1.1 T and cooled through the isotropic–hexagonal transition, both the nanoparticles and the hexagonal domains align parallel to the magnetic field. Although the transport properties are found to be drug-specific, in the case of glucose diffusion experiments it was shown that the hexagonal phase aligned horizontally and parallel to the diffusion chamber ($D_{||}$) shows more than an 8-fold increase in the diffusion coefficient compared to that of the hexagonal phase aligned in the vertical direction ($D_{\perp}$). This clearly demonstrates that in the case of anisotropic mesophases not only the size of water domains and the symmetry but also the orientation of the domains plays a role in the diffusion. Additionally, this work shows that specific drug–mesophase interactions play a critical role in determining the diffusivity, even in the aligned domains. The good biocompatibility of lipid-based lyotropic liquid crystals and the Fe_3O_4 magnetic nanoparticles with noninvasive programmable anisotropy in transport properties can serve in bioapplications such as drug delivery, biosensors, and biomedical materials.

ASSOCIATED CONTENT

Supporting Information

Experimental procedures, synthesis of magnetic nanoparticles, and methods. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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